

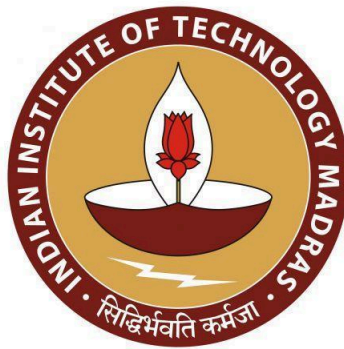
Software Engineering Project May-2026

Milestone 4 Deliverables for Campus Connect

Submitted By

Team Name: NexMind

Team Code : Team-003



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Student Club & Event Management Platform for Higher-Education Campuses

Project Details

Field	Value
Software Name	Campus Connect
Institutional Context	Campus Council / Student Club Ecosystem
System Type	Community Operations Platform (Club, Event & Participation Management)
Milestone	Milestone 4 - Sprint 2: Complete API Implementation, Testing, Integration and User Feedback
Sprint Window	3 August 2026 - 12 August 2026
Date of Submission	12-08-2026
Live API Base URL	https://campusconnect.itshrestha.dev
Interactive API Documentation	https://campusconnect.itshrestha.dev/docs (Swagger UI)
Machine-readable Specification	https://campusconnect.itshrestha.dev/openapi.json
Frontend Deployment	https://campus-connect-swe.vercel.app/
Code Repository	https://github.com/Srivastava-Shrestha/MAY2026-Team-003

Team Details

Team Name: NexMind **Team Code:** Team-003

Name	Roll Number	Role in the Project
Atharv Khare	23f2004201	Lead Product Manager / Frontend Developer
Kavisha Tankle	23f1000041	Scrum Master / QA Engineer
Shrestha Srivastava	23f3000168	Backend Lead / Code Reviewer
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Shrishti Gupta	23f2004336	Frontend Developer

Contents

1. Sprint 2 Overview and Scope
2. Milestone 4 Task Distribution
3. Changes Made Based on Sprint 1 User Feedback
4. API Design Updates and the Status of DOC-01
5. List of APIs Created and Updated
6. List of APIs and Services Integrated
7. Updated API to User-Story Traceability Matrix
8. Test Cases
9. Unit Testing with pytest
10. Defects Found – Expected vs Actual
11. Frontend and Backend Integration
12. User Feedback from the Sprint 2 Demo
13. Plan for Milestone 5 (Final Submission)
14. Submission Artefacts

1. Sprint 2 Overview and Scope

1.1 Sprint Goal

Sprint 1 delivered the participation spine of Campus Connect: publish an event, register, check in, record a result. Sprint 2 delivers what hangs off the far end of that spine. The certificate engine (Stories 5.1-5.3) now fires automatically when a leader declares results, so a participation record becomes a downloadable, publicly verifiable PDF without any further human action. Password recovery was added in direct response to the Sprint 1 demo, and the result-declaration workflow was rebuilt around the way club leaders described actually running an event.

Two committed stories did not ship this sprint and are reported as such rather than presented as partially done. The AI Club Finder (Story 6.1) was rebuilt as an agentic module after client and TA feedback and is complete in isolation, but integrating it now would force the rest of the system to accommodate an architecture that is still settling; it is therefore scheduled for Milestone 5. The club activity leaderboard (Story 7.1) is likewise carried forward. As in Sprint 1, nothing outside the Milestone 1 backlog was built – every endpoint traces to a committed story or a numbered feedback item.

1.2 Sprint Window and Cadence

Item	Detail
Sprint 2 window	3 August 2026 – 12 August 2026 (10 days)
Target release	Milestone 4 – 12 August 2026
Stand-up	Daily asynchronous status collection at 9:00 AM and 9:00 PM on the team channel; a live stand-up on Google Meet whenever a blocker is raised
Sprint planning	3 August 2026 – Sprint 1 review outcomes converted into the Sprint 2 backlog with ownership confirmation
Sprint review / client demo	Client window from 16 August 2026 (exact date to be confirmed)
Sprint retrospective	13 – 14 August 2026 (internal team review after submission)
Tracking board	Trello – https://trello.com/b/M4qafINT/nexmind

1.3 Stories Committed in Sprint 2

Story	Title	Module	Endpoints	Status
5.1	Auto-generate participation certificates	Certificate	Triggered in the background by PATCH /events/{event_id}/results	Delivered
5.2	Download my participation certificate	Certificate	GET /certificates/me; GET /certificates/{serial}/download	Delivered
5.3	Verify a certificate by serial number	Certificate	GET /certificates/verify/{serial}	Delivered
6.1	Find clubs by describing my interests (AI Club Finder)	AI Discovery	–	Carried to Milestone 5

Story	Title	Module	Endpoints	Status
7.1	Club activity leaderboard	Recognition	-	Carried to Milestone 5

Story 6.1 was carried into Sprint 2 from Sprint 1. Following client and TA feedback the deterministic AI club recommender was redesigned as an agentic module. That module is complete and tested in isolation, but it is not yet integrated: bolting it on at the end of Sprint 2 would have meant shipping an integration the team could not test to the same standard as the rest of the system. It is reported as carried over rather than shown as delivered, and is scheduled for Milestone 5 alongside the remaining frontend integration work.

1.4 Feedback Items from the Sprint 1 Demo

Two items carried in the Milestone 3 backlog – a written reason on club rejection and an expiry on pinned announcements – were withdrawn after review rather than deferred, so the remaining items are renumbered contiguously here in their original Milestone 3 order. F-06 is new: password recovery was raised by a primary user at the Sprint 1 demo and recorded in the Milestone 3 feedback narrative, but was not given an item id at the time.

ID	Source	Change requested	Owner	Status
F-01	Club leader	Bulk check-in and a searchable participant roster on the attendance console	Pawan / Shrishti	Carried to M5
F-02	Club leader	Default every checked-in attendee to PARTICIPANT with per-student override for WINNER and RUNNER_UP	Shrestha	Delivered
F-03	Student	Event reminder notification	Shrestha	Delivered
F-04	Ambassador	Seen count on an announcement for the leader dashboard	Pawan	Carried to M5
F-05	Team (DOC-01)	Declare 401 / 403 / 404 / 409 responses and story ids on every route decorator and re-export the YAML	Shrestha	In progress - see §4.3
F-06	Student (new)	Forgot-password and reset-password recovery	Shrestha	Delivered

1.5 Sprint 2 Outcome at a Glance

Metric	Value
User stories committed	5 (Stories 5.1, 5.2, 5.3, 6.1, 7.1)
User stories delivered	3 (5.1, 5.2, 5.3); 6.1 and 7.1 carried to Milestone 5
Feedback items delivered	3 of 6 (F-02, F-03, F-06); F-05 in progress; F-01 and F-04 carried to Milestone 5
New API operations created in Sprint 2	5 (2 Auth + 3 Certificate)
API operations replaced	1 (PATCH /events/{event_id}/registrations/{registration_id}/result superseded by PATCH /events/{event_id}/results)

Metric	Value
Total API operations (cumulative)	50 operations across 42 distinct resource paths
Functional modules covered	7 of 8 delivered (Identity & Access, Club Management, Membership, Event Management, Communication, Notifications, Certificate); AI Discovery and Recognition carried to M5
Test cases in the cumulative suite	459 (426 integration, IT-001 - IT-426; 33 unit, UT-427 - UT-459)
Test cases added in Sprint 2	60 (15 certificate integration, 12 password-reset integration, 33 unit)
pytest cases executed	459
Defects found where expected $\neq$ actual	0 functional in Sprint 2; DOC-01 (documentation) remains open from Milestone 3 - see §10

1.6 Milestone 4 Rubric Coverage

Rubric requirement	Section	Owner
Changes made based on user feedback	§3	Atharv, Kavisha
Updated Swagger-compatible YAML documentation	§4.3 + campus-connect-openapi.yaml	Shrestha
Backend code of all implemented APIs	Submitted zip - backend/	Shrestha, Pawan
Test cases (API, inputs, expected, actual, result)	§8	Pawan, Kavisha
Snapshot of pytest functions	§8.5	Pawan
pytest code and output	§9 + backend/tests/	Pawan
Case where expected $\neq$ actual	§10	Pawan, Kavisha
User feedback on the delivered features	§12	Atharv, Kavisha
Plan for the next sprint	§13	Atharv, Kavisha
Facilitating and composing Milestone 4 deliverables	-	Atharv, Kavisha

2. Milestone 4 Task Distribution

Roles carried unchanged from Sprint 1. The table records what each member owned during Sprint 2 and which artefact in this submission they are accountable for. Work on a story that was ultimately carried to Milestone 5 is recorded as such rather than credited as delivered.

Member	Role	Sprint 2 work owned	Milestone 4 artefact owned
Atharv Khare	Lead Product Manager / Frontend Developer	Sprint planning and milestone tracking; redesign of the AI Club Finder into an agentic module after client and TA feedback, built and tested in isolation and scheduled for integration in Milestone 5; user acceptance testing of every delivered story; product quality sign-off before merge to main; running the Sprint 2 demo	This report (assembly and final review), §1, §3, §12, §13
Kavisha Tangle	Scrum Master / QA Engineer	Board hygiene on Trello and blocker resolution; facilitated the daily 9 AM stand-ups; drafted minutes for client and internal meetings; pre-acceptance validation of the certificate and password-recovery flows against Swagger before UAT; tracking the team's GitHub activity (pull requests, issues, branches)	Scrum minutes, GitHub issue records, §8 test-case matrix review, §10 defect log
Shrestha Srivastava	Backend Lead / Code Reviewer	Certificate engine - SVG to PDF rendering, QR embedding, serial generation and the public verification endpoint; the rebuilt single-call result declaration endpoint (F-02); forgot-password and reset-password endpoints with Resend delivery (F-06); review and merge of every pull request	§4 API design updates, §5 endpoint catalogue, campus-connect-openapi.yaml, backend/ source
Pawan Kumar Choudhary	Backend Developer / Tester	Certificate integration and unit test suites; password-reset test suite including token misuse and expiry paths; regression coverage across every Sprint 1 endpoint touched by the result-declaration rework; maintenance of the cumulative 459-case matrix	§6 integrated services, §8 test cases, §9 pytest suite and screenshots, backend/tests/
Shrishti Gupta	Frontend Developer	Replacing the remaining frontend mocks with live calls across Authentication, Onboarding, Clubs, Events and Announcements; password-recovery screens; token refresh and route guards; loading, empty and error states on every wired view	§11 integration status

3. Changes Made Based on Sprint 1 User Feedback

The course requirement for Sprint 2 is to show the changes made in response to user feedback, so this section is organised by feedback item rather than by module. Each item states what the user observed at the Sprint 1 demo, what changed, and where the change lives in the API. Items that were not delivered are reported here as not delivered, with the reason, rather than being quietly omitted.

3.1 F-01 Bulk Check-in and Searchable Roster – carried to Milestone 5

Not delivered in Sprint 2. The attendance console still marks one registration at a time through `PATCH /events/{event_id}/registrations/{registration_id}/attendance`. The item stays in the backlog for Milestone 5; the sprint's capacity went to the certificate engine and to the result-declaration rework that the same user requested alongside it.

3.2 F-02 Default-and-Override Result Flow – delivered

The club leader's observation at the Sprint 1 demo was that setting results one attendee at a time does not match how an event actually ends: everyone who turned up is a participant, and then there are two exceptions. Sprint 1 exposed a per-registration endpoint that required one call per attendee.

That endpoint was replaced by a single call. `PATCH /events/{event_id}/results` accepts `winner_registration_id` and `runner_up_registration_id`, assigns `PARTICIPANT` to every other checked-in attendee, queues certificates for all of them, and then freezes both results and attendance for that event so the record cannot drift afterwards. The response reports the winner, the runner-up, the participant count and the number of certificates queued.

- Attempting to declare results twice returns 409 with an explicit message that results have already been declared and can no longer be changed (IT-279).
- Marking attendance after declaration returns 409 on the same guard (IT-283, IT-284).
- A registration that was never checked in cannot receive a result: 400 (IT-277).

3.3 F-03 Event Reminder Notification – delivered

A primary user reported at the Sprint 1 demo that the registration confirmation is easy to lose in the feed and that nothing prompts her again as the event approaches. Event reminder notifications were implemented in Sprint 2, so a student who has registered is prompted ahead of the event rather than having to go back and find the original confirmation.

3.4 F-04 Announcement Seen Count – carried to Milestone 5

Not delivered in Sprint 2. `GET /announcements/unread-count` returns the number of announcements the caller has not yet seen, which serves the student's navbar badge; it does not answer the ambassador's question, which was how many members have read a

given post. That requires per-member read tracking on an announcement rather than a single seen-at marker per user, and it is carried to Milestone 5 as a schema change rather than reported as satisfied by an endpoint that measures something else.

3.5 F-05 / DOC-01 Specification Error Responses – in progress

DOC-01 was logged in Milestone 3: the generated specification declares only 200 and 422 for each operation, because non-2xx paths are raised as HTTPException without a responses= mapping on the route decorator. The behaviour is correct at runtime and is evidenced by the test cases in §8 – 401, 403, 404 and 409 are all asserted against the deployed build – but the specification does not yet document them, and the per-operation descriptions do not yet carry their user story ids.

The work is in progress and is not claimed as complete in this submission. It is carried to Milestone 5 and tracked in §10 and §13.1. See §4.3 for the current state of the specification and what remains.

3.6 F-06 Forgot Password and Reset Password – delivered

A primary user asked at the Sprint 1 demo for Google authentication and a forgot-password feature. Password recovery was implemented in Sprint 2 as two endpoints: POST /auth/forgot-password issues a single-use, time-limited reset token and emails it via Resend, and POST /auth/reset-password consumes that token to set a new password.

- The forgot-password response is deliberately identical whether or not the email is registered, so the endpoint cannot be used to enumerate accounts (IT-415, IT-416).
- A reset token is single-use: a second attempt with the same token returns 401 (IT-420).
- Access and refresh tokens are rejected as reset tokens, as are expired and tampered tokens (IT-421 to IT-424).
- After a successful reset, the old password no longer authenticates and the new one does (IT-419).

Google authentication was the second half of that request. It is recorded in the future-scope list with its reason: it introduces an external identity provider and a second account-linking path, which is a subsystem of its own rather than an increment on an existing entity, and the team has applied that scope test consistently since Milestone 1.

4. API Design Updates and the Status of DOC-01

4.1 What Changed Since Milestone 3

Three changes were made to the API surface between Milestone 3 and Milestone 4.

- The Certificate module was implemented: three endpoints covering the student certificate wallet, a fresh time-limited download link, and public serial verification. Certificate creation itself is not an endpoint – it is a background consequence of declaring results, which is what Story 5.1 asks for.
- Two Auth endpoints were added for password recovery (F-06).
- The per-registration result endpoint was replaced by a single event-level declaration endpoint (F-02). This is a replacement rather than an addition, which is why the cumulative operation count moves from 45 to 50 rather than to 51.

4.2 Schema Changes

One new table was added: Certificate, storing the serial number, the student and event it belongs to, the result it records, the issue timestamp and the storage reference for the rendered PDF. No new enumerations were introduced – RegistrationResult (REGISTRANT, PARTICIPANT, RUNNER_UP, WINNER), already in place from Sprint 1, drives certificate eligibility, and CERTIFICATE_ISSUED already existed in NotificationType.

Certificate serials follow the format CC-**<EVENT-ABBREVIATION>**-**<YEAR>**-**<SEQUENCE>**, for example CC-LFW-2026-00001. The abbreviation is derived from the event title, stripped of punctuation and clipped to a fixed width; the sequence is allocated per college per year. The derivation is covered by unit tests UT-427 to UT-436.

4.3 Swagger-Compatible YAML Documentation – current state

The specification is generated by FastAPI from the Pydantic request and response models, so it cannot drift from the implementation, and is exported to YAML for submission with:

```
curl -s https://campusconnect.itshrestha.dev/openapi.json | python -c "import sys,json,yaml;yaml.safe_dump(json.load(sys.stdin),sys.stdout,sort_keys=False)" > campus-connect-openapi.yaml
```

The submitted file is campus-connect-openapi.yaml (OpenAPI 3.1.0, Swagger-compatible). It documents all 50 operations across 42 paths, the HTTPBearer security scheme with per-operation role scopes, and every request and response schema. The same document renders as the live Swagger UI at /docs, which the QA lead used for manual lifecycle validation of the certificate and password-recovery flows.

[Link for the live Swagger UI screenshot](#)

[Link for the live Swagger website](#)

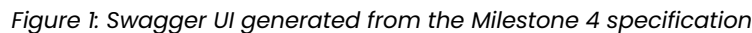




Figure 2: Sample participation certificate generated by the Campus Connect certificate engine ([Link to the pdf formatted sample](#))

5. List of APIs Created and Updated

The five operations below were designed and implemented by the team in Sprint 2, continuing the numbering from the Milestone 3 catalogue. The Access column gives the platform role required by the route guard; where a route is marked STUDENT but the action is a leader action, the membership-level check described in Milestone 3 §3.2 applies inside the handler.

5.1 Certificates (3 operations)

#	Method	Endpoint	Description	Access	Story
46	GET	/certificates/me	Returns the caller's certificate wallet – every certificate issued to them, newest first, each carrying the event it came from, the club that ran it, the result it records, the issue timestamp and a ready-to-use download link.	STUDENT	5.2
47	GET	/certificates/{serial}/download	Issues a fresh, time-limited signed URL (expires_in 3600) to one certificate PDF. Does not return the PDF directly – the client follows the link to object storage. Restricted to the owning student; a certificate belonging to anyone else answers 404 rather than 403, so ownership is not leaked.	STUDENT	5.2
48	GET	/certificates/verify/{serial}	Public endpoint – no token required. Resolves a serial to the record behind it: who earned it, for which event, at which college, under which club, with what result and when it was issued. This is the page behind the QR code printed on every certificate.	Public	5.3

5.2 Auth – password recovery (2 operations)

#	Method	Endpoint	Description	Access	Story
49	POST	/auth/forgot-password	Issues a single-use, time-limited reset token for a registered address and emails it via Resend. The response is identical for registered and unregistered addresses so the endpoint cannot be used to enumerate accounts.	Public	F-06
50	POST	/auth/reset-password	Consumes a reset token and sets a new password. Rejects reused, expired, malformed and tampered tokens, and rejects access or refresh tokens presented in their place, all with 401.	Public	F-06

5.3 Operation Replaced in Sprint 2

#	Method	Endpoint	What changed	Access	Story
28	PATCH	/events/{event_id}/results	Replaces PATCH /events/{event_id}/registrations/{registration_id}/result from Sprint 1. Declares the winner and the runner-up in one call; every other checked-in attendee stays PARTICIPANT; certificates are generated for all of them in the background; results and attendance are frozen afterwards. Implements F-02.	STUDENT (club LEADER)	3.5, 5.1

5.4 Modules Not Yet Implemented

No AI Discovery or Recognition endpoints appear in the Milestone 4 specification. Story 6.1 and Story 7.1 are carried to Milestone 5 for the reasons recorded in §1.3 and §13. They are listed here explicitly so the endpoint catalogue and the traceability matrix in §7 agree with each other.

6. List of APIs and Services Integrated

In addition to the endpoints created by the team, Campus Connect consumes the third-party services below. Nothing in this list replaces team-written business logic; each is used for a single, well-defined capability.

Service / API	Purpose	Where it is used	Status in Sprint 2
Neon serverless PostgreSQL	Managed relational store for every entity in the ER design	All backend modules, via SQLAlchemy	Live
PyJWT	Signing and verification of access, refresh and password-reset tokens	Auth endpoints and every protected route	Live
cairosvg + Jinja2 + qrcode[pil]	SVG-to-PDF certificate rendering with an embedded verification QR code	Certificate module, Stories 5.1-5.3	Live
Resend Email API	Transactional email	Password-reset link delivery (F-06); signup confirmation	Live
MailHog	Local SMTP capture for testing email delivery without sending real mail	Password-reset test suite (IT-415, IT-416)	Live (test environment)
Anthropic Messages API (Claude Haiku)	Free-text reason strings for the AI Club Finder; ranking stays in our own code	Story 6.1 - thin backend proxy so the API key is never exposed to the browser	Carried to Milestone 5*
Render.com	Backend hosting and automatic deploys from the main branch	Deployment pipeline	Live
Vercel	Frontend hosting and preview deployments per branch	Deployment pipeline	Live

**The AI Club Finder remains the only GenAI integration in the product, and the key stays server-side behind our own proxy endpoint. Following client and TA feedback the module was redesigned from a deterministic scorer with model-written reasons into an agentic module. That redesign is complete and tested in isolation; integration is scheduled for Milestone 5 so it can be validated to the same standard as the rest of the system.*

7. Updated API to User-Story Traceability Matrix

The matrix is read story-first, so that no committed story is left without an implementation and no endpoint exists without a story behind it. It spans both sprints: Sprint 1 rows are carried forward with their endpoints, and Sprint 2 rows are completed here.

Test-case identifiers were renumbered for Milestone 4 when the cumulative matrix was reorganised by module. The ranges below are the Milestone 4 numbering and supersede the ranges published in Milestone 3 §6.

Story	User story title	Endpoints implementing it	Owner	Status	Test cases
1.1	Register a college on the platform	POST /college/onboarding	Shrestha	Delivered	IT-187 - IT-208
1.2	Sign up and log in	POST /auth/signup; POST /auth/login	Shrestha	Delivered	IT-057 - IT-083
2.1	Browse the club directory	GET /clubs	Shrestha	Delivered	IT-123 - IT-133
2.2	View a club profile	GET /clubs/{club_id}; GET /clubs/{club_id}/members	Shrestha	Delivered	IT-134 - IT-140, IT-152 - IT-153, IT-373 - IT-377
2.3	Request to join a club	POST /clubs/{club_id}/join; GET /clubs/me	Shrestha	Delivered	IT-353 - IT-359, IT-175 - IT-186
2.4	Manage membership join requests	GET /clubs/{club_id}/requests; PATCH /clubs/{club_id}/requests/{membership_id}	Shrestha	Delivered	IT-360 - IT-372
2.5	Create a club and request approval	POST /clubs; PUT /clubs/{club_id}; DELETE /clubs/{club_id}	Shrestha	Delivered	IT-099 - IT-122, IT-141 - IT-163
2.6	Approve or reject club registration	PATCH /clubs/{club_id}/approve; PATCH /clubs/{club_id}/reject	Shrestha	Delivered	IT-164 - IT-174
3.1	Create and publish an event	POST /events; PATCH /events/{event_id}/publish; GET /events	Shrestha	Delivered	IT-209 - IT-222, IT-235 - IT-243
3.2	Register for an event with confirmation	GET /events/{event_id}; POST /events/{event_id}/register; DELETE /events/{event_id}/register; GET /events/me/registrations	Shrestha	Delivered	IT-244 - IT-260, IT-269 - IT-271
3.3	View registered participants	GET /events/{event_id}/registrations	Shrestha	Delivered	IT-261 - IT-264

Story	User story title	Endpoints implementing it	Owner	Status	Test cases
3.4	Mark attendance at an event	PATCH /events/{event_id}/registrations/{registration_id}/attendance	Shrestha	Delivered	IT-265 - IT-268, IT-283 - IT-284
3.5	Set a result for each attendee	PATCH /events/{event_id}/results (reworked in Sprint 2 for F-02)	Shrestha	Delivered	IT-275 - IT-282, IT-285 - IT-286
3.6	View my event result	GET /events/me/results	Shrestha	Delivered	IT-272 - IT-274
3.7	Edit or cancel an event	PUT /events/{event_id}; PATCH /events/{event_id}/cancel	Shrestha	Delivered	IT-223 - IT-234
4.1	Post a club announcement	POST /announcements; GET /announcements/mine; PATCH /announcements/{id}/pin; DELETE /announcements/{id}	Shrestha	Delivered	IT-001 - IT-016, IT-028 - IT-030, IT-035 - IT-045, IT-047, IT-055 - IT-056
4.2	View announcements from joined clubs	GET /announcements; GET /announcements/unread-count; POST /announcements/read-all	Shrestha	Delivered	IT-017 - IT-027, IT-031 - IT-034, IT-046, IT-048 - IT-054
4.3	Raise an issue or query	POST /issues; GET /issues	Shrestha	Delivered	IT-287 - IT-314
4.4	View and resolve member issues	GET /issues/club; GET /issues/club/open-count; POST /issues/{id}/reply; PATCH /issues/{id}/resolve	Shrestha	Delivered	IT-315 - IT-352
2.4, 3.2, 3.5	Notification inbox supporting the above stories	GET /notifications; GET /notifications/unread-count; POST /notifications/read-all; PATCH /notifications/{id}/read	Shrestha	Delivered	IT-378 - IT-414
5.1	Auto-generate participation certificates	No separate endpoint - generated in the background by PATCH /events/{event_id}/results	Shrestha	Delivered (Sprint 2)	IT-275, IT-286; UT-437 - UT-444
5.2	Download my participation certificate	GET /certificates/me; GET /certificates/{serial}/download	Shrestha	Delivered (Sprint 2)	IT-084 - IT-087, IT-090 - IT-095, IT-098; UT-427 - UT-436
5.3	Verify a certificate by serial number	GET /certificates/verify/{serial}	Shrestha	Delivered (Sprint 2)	IT-088 - IT-089, IT-096 - IT-097
F-06	Recover a forgotten password	POST /auth/forgot-password; POST /auth/reset-password	Shrestha	Delivered (Sprint 2)	IT-415 - IT-426; UT-445 - UT-459
6.1	Find clubs by describing my interests (AI Club Finder)	Not yet integrated - agentic module complete in isolation	Atharv	Carried to M5	-

Story	User story title	Endpoints implementing it	Owner	Status	Test cases
7.1	Club activity leaderboard	Not yet implemented	Shrestha	Carried to M5	-

8. Test Cases

Test cases are written in the format required by the course – API being tested, inputs, expected output, actual output and result. The cumulative suite stands at **459 cases**: 426 integration cases (IT-001 to IT-426) and 33 unit cases (UT-427 to UT-459). 60 of these were added in Sprint 2. The Sprint 2 sections are reproduced in full below; the complete matrix is available at the link.

Full test-case document: [Campus Connect – complete API test case matrix \(459 cases\)](#)

The Actual Output column records the observed behaviour of the deployed build at the end of Sprint 2. Every case in the suite passed on the run recorded in §9.3; §10 discusses what that does and does not tell us.

8.1 Certificates – Stories 5.1 to 5.3 (15 cases, IT-084 to IT-098)

ID	API being tested	Inputs	Expected Output	Actual Output	Result
IT-084	GET /certificates/me	Student with one earned certificate	200; len==1; serial 'CC-LFW-2026-00001'; result 'WINNER'; 'download_url' present in body	matches expected	Success
IT-085	GET /certificates/me	Student with zero certificates	200; []	matches expected	Success
IT-086	GET /certificates/me	No token	401; detail 'Not authenticated'	matches expected	Success
IT-087	GET /certificates/me	Non-student caller	401	matches expected	Success
IT-088	GET /certificates/verify/CC-LFW-2026-00001	No auth headers	200; valid True; serial matches; result 'WINNER'	matches expected	Success
IT-089	GET /certificates/verify/CC-FAKE-9999-00000	Serial that does not exist	404; msg 'Certificate not found'	matches expected	Success
IT-090	GET /certificates/{serial}/download	Owning student	200; serial; filename; download_url; expires_in 3600	matches expected	Success

ID	API being tested	Inputs	Expected Output	Actual Output	Result
IT-091	GET /certificates/{serial}/download	Non-owning student	404; msg 'Certificate not found'	matches expected	Success
IT-092	GET /certificates/FAKE-SERIAL/download	Serial that does not exist	404; msg 'Certificate not found'	matches expected	Success
IT-093	GET /certificates/{serial}/download	No token	401; detail 'Not authenticated'	matches expected	Success
IT-094	GET /certificates/me	Certificate from a single event	200; len==1; serial 'CC-LFW-2026-00001'	matches expected	Success
IT-095	GET /certificates/me	Certificates from two events	200; len==2; both serials present	matches expected	Success
IT-096	GET /certificates/verify/CC-LFW-2026-00050	Certificate recording a RUNNER_UP result	200; result 'RUNNER_UP'	matches expected	Success
IT-097	GET /certificates/verify/CC-LFW-2026-00099	Certificate recording a PARTICIPANT result	200; result 'PARTICIPANT'	matches expected	Success
IT-098	GET /certificates/{serial}/download	RUNNER_UP certificate, owning student	200; serial 'CC-LFW-2026-00051'	matches expected	Success

8.2 Password Recovery – F-06 (12 cases, IT-415 to IT-426)

ID	API being tested	Inputs	Expected Output	Actual Output	Result
IT-415	POST /auth/forgot-password	{"email": "itsmepawan.jh@gmail.com"} (registered address)	200; msg 'If that email is registered, a password reset link has been sent to it'; reset mail captured	matches expected	Success
IT-416	POST /auth/forgot-password (MailHog inspected)	Unregistered email address	200; identical generic message; 0 items captured in MailHog	matches expected	Success
IT-417	POST /auth/forgot-password	{"email": "not-an-email"}	422	matches expected	Success
IT-418	POST /auth/forgot-password	{}	422	matches expected	Success
IT-419	POST /auth/reset-password then POST /auth/login	Valid reset, then login with the old password and with the new one	200 on reset; old-password login 401 'Incorrect email or password'; new-password login 200 with tokens	matches expected	Success
IT-420	POST /auth/reset-password	Same reset token submitted twice	First call 200; second call 401 'Invalid token'	matches expected	Success
IT-421	POST /auth/reset-password	Expired reset token	401; msg 'Invalid token'	matches expected	Success
IT-422	POST /auth/reset-password	Access token presented as a reset token	401; msg 'Invalid token'	matches expected	Success
IT-423	POST /auth/reset-password	Refresh token presented as a reset token	401; msg 'Invalid token'	matches expected	Success
IT-424	POST /auth/reset-password	Malformed / tampered token	401; msg 'Invalid token'	matches expected	Success
IT-425	POST /auth/reset-password	password and confirm_password do not match	422	matches expected	Success
IT-426	POST /auth/reset-password	Password below the minimum length	422	matches expected	Success

8.3 Result Declaration and Certificate Trigger - F-02, Stories 3.5 and 5.1 (12 cases, IT-275 to IT-286)

These are the regression cases for the Sprint 1 endpoint replaced this sprint. They also stand as the evidence for Story 5.1, since certificate generation has no endpoint of its own and is observed through certificates_queued on this response.

ID	API being tested	Inputs	Expected Output	Actual Output	Result
IT-275	PATCH /events/{event_id}/results	{'winner_registration_id': first_id, 'runner_up_registration_id': second_id}	200; winner and runner_up set; participants 0; certificates_queued 2; msg 'Results declared, certificates are being generated'	matches expected	Success
IT-276	PATCH /events/{event_id}/results	winner_registration_id == runner_up_registration_id	422	matches expected	Success
IT-277	PATCH /events/{event_id}/results	Registration that was never checked in	400; msg 'Result can only be set for attendees who were checked in'	matches expected	Success
IT-278	PATCH /events/{event_id}/results	Registration id nonexistent or belonging to another event	404; msg 'Registration not found'	matches expected	Success
IT-279	PATCH /events/{event_id}/results	Declaring results a second time	409; msg 'Results for this event have already been declared and can no longer be changed'	matches expected	Success
IT-280	PATCH /events/{event_id}/results	Caller is a member, not the club leader	403	matches expected	Success
IT-281	PATCH /events/999999/results	Event that does not exist	404; msg 'Event not found'	matches expected	Success
IT-282	PATCH /events/{event_id}/results	No token	401; detail 'Not authenticated'	matches expected	Success
IT-283	PATCH /events/{event_id}/registrations/{registration_id}/attendance	Marking attendance after results were declared	409; msg 'Results for this event have already been declared and can no longer be changed'	matches expected	Success
IT-284	PATCH /events/{event_id}/registrations/{registration_id}/attendance	Attendance change on a non-winner / non-runner-up after the freeze	409	matches expected	Success
IT-285	PATCH /events/{other_event_id}/results	Registration id belonging to a different event	404; msg 'Registration not found'	matches expected	Success

ID	API being tested	Inputs	Expected Output	Actual Output	Result
IT-286	PATCH /events/{event_id}/results	Five checked-in attendees; participants count must exclude winner and runner-up	200; participants 1; certificates_queued 3	matches expected	Success

8.4 Unit Tests – certificate rendering, serials and tokens (selection from UT-427 to UT-459)

ID	Function under test	Inputs	Expected Output	Actual Output	Result
UT-427	abbreviate()	'Data Science & Analytics'	returns 'DSA'	matches expected	Success
UT-429	abbreviate()	'12345'	returns '1' – digits do not contribute initials	matches expected	Success
UT-430	abbreviate()	'---' (no alphabetic content)	falls back to 'EVT'	matches expected	Success
UT-431	abbreviate()	'C++ & Python: Bootcamp'	result.isalpha() and length within the fixed width	matches expected	Success
UT-432	make_serial()	'Robotics Workshop', year 2026	matches ^CC-RW-2026-\d{5}\$	matches expected	Success
UT-433	make_serial()	'---', year 2026	matches ^CC-EVT-2026-\d{5}\$	matches expected	Success
UT-434	clip()	'Very long event title...', width 15	returns 'Very long...' with an ellipsis	matches expected	Success
UT-437	render_certificate() – WINNER	CertificateContext for a WINNER result	pdf_bytes.startswith(b'PDF')	matches expected	Success
UT-440	CertificateContext()	Very long student and event names	pdf_bytes.startswith(b'PDF') – long fields are clipped, not overflowed	matches expected	Success
UT-441	issue() with a failing serial allocator	AsyncMock raising on serial allocation	raises RuntimeError – the failure is not swallowed	matches expected	Success
UT-442	issue() for a registration never checked in	MagicMock registration, checked_in False	result is None – no certificate is issued	matches expected	Success
UT-445	hash_password()	'MyPassword123'	hashed value differs from the plaintext	matches expected	Success
UT-447	verify_password()	'WrongPassword' against a stored hash	returns False	matches expected	Success

ID	Function under test	Inputs	Expected Output	Actual Output	Result
UT-455	create_refresh_token()	Payload missing a required claim	raises	matches expected	Success
UT-457	create_reset_token()	{'sub': '1', ...}	type == 'reset'; sub and expiry present	matches expected	Success

8.5 Snapshots of pytest Functions

The snapshots below are taken from the Sprint 2 suites. Two cover the Certificate module delivered this sprint; two cover the Notification module, where the regression risk was highest because the result-declaration rework changed when and how notifications are raised.

```
@pytest.mark.asyncio
async def test_my_certificates_lists_multiple_events_same_student(client, db_session, leader, member):
    """Verify that a student's certificates from two different events both appear in their list"""
    leader_headers, club_id = leader

    events = []
    for i in range(2):
        payload = {
            "club_id": club_id,
            "title": f"Workshop {i}",
            "description": f"Test workshop number {i} for multi-certificate testing",
            "venue": "Lab 204",
            "starts_at": future_time(2 + i * 10),
            "ends_at": future_time(4 + i * 10),
        }
        create = await client.post("/events", headers=leader_headers, json=payload)
        event_id = create.json()["id"]
        await client.patch(f"/events/{event_id}/publish", headers=leader_headers)
        registration = await client.post(f"/events/{event_id}/register", headers=member)
        events.append((event_id, registration.json()["registration_id"]))

    for i, (event_id, registration_id) in enumerate(events):
        certificate = Certificate(
            registration_id=registration_id,
            serial=f"CC-MULTI-2026-0000{i+1}",
            result=RegistrationResult.PARTICIPANT,
            issued_at=datetime.now(timezone.utc),
```

Figure 3: test_my_certificates_lists_multiple_events_same_student - a student who earns certificates at two separate events sees both in their wallet (IT-095)

```

235
236
237 @pytest.mark.asyncio
238 async def test_download_certificate_by_non_owner_fails(client, certificate_for_member, outsider):
239     """Ensure that a student who does not own the certificate cannot download it"""
240     _, _, certificate = certificate_for_member
241     response = await client.get(f"/certificates/{certificate.serial}/download", headers=outsider)
242     assert response.status_code == 404
243     assert response.json()["message"] == "Certificate not found"
244
245
246 @pytest.mark.asyncio
247 async def test_download_certificate_unknown_serial_fails(client, member):
248     """Confirm that downloading a non-existent serial returns not found"""
249     response = await client.get("/certificates/FAKE-SERIAL/download", headers=member)
250     assert response.status_code == 404
251     assert response.json()["message"] == "Certificate not found"
252
253

```

Figure 4: `test_download_certificate_by_non_owner_fails` and `test_download_certificate_unknown_serial_fails` - a non-owner and an unknown serial both answer 404, so ownership is never leaked (IT-091, IT-092)

```

9 @pytest.mark.asyncio
10 async def test_unread_count_decrements_by_one_after_single_read(client, leader, member):
11     """Verify that marking a single notification as read decrements the unread count by exactly 1"""
12     headers, club_id = leader
13     payload = {
14         "club_id": club_id,
15         "title": "Single decrement event",
16         "description": "An event used to test single-notification unread count decrement",
17         "venue": "Lab 204",
18         "starts_at": future_time(48),
19         "ends_at": future_time(50),
20     }
21     create = await client.post("/events", headers=headers, json=payload)
22     event_id = create.json()["id"]
23     await client.patch(f"/events/{event_id}/publish", headers=headers)
24     await client.post(f"/events/{event_id}/register", headers=member)
25
26     before = await client.get("/notifications/unread-count", headers=member)
27     initial_count = before.json()["count"]
28     assert initial_count >= 1
29
30     listed = await client.get("/notifications", headers=member)
31     notification_id = listed.json()[0]["id"]
32     await client.patch(f"/notifications/{notification_id}/read", headers=member)
33
34     after = await client.get("/notifications/unread-count", headers=member)
35     assert after.json()["count"] == initial_count - 1
36

```

Figure 5: test_unread_count_decrements_by_one_after_single_read - marking one notification read moves the badge by exactly one, not to zero

```

@pytest.mark.asyncio
async def test_unregister_does_not_remove_earlier_registration_notification(client, leader, member):
    """Confirm that cancelling a registration does not delete the earlier REGISTRATION_CONFIRMED notification"""
    headers, club_id = leader
    payload = {
        "club_id": club_id,
        "title": "Unregister notification test",
        "description": "An event used to test notification persistence after unregistering",
        "venue": "Lab 204",
        "starts_at": future_time(48),
        "ends_at": future_time(50),
    }
    create = await client.post("/events", headers=headers, json=payload)
    event_id = create.json()["id"]
    await client.patch(f"/events/{event_id}/publish", headers=headers)
    await client.post(f"/events/{event_id}/register", headers=member)
    await client.delete(f"/events/{event_id}/register", headers=member)

    response = await client.get(
        "/notifications", headers=member, params={"type": "REGISTRATION_CONFIRMED"}
    )
    assert response.status_code == 200
    assert len(response.json()) == 1

```

Figure 6: `test_unregister_does_not_remove_earlier_registration_notification` - withdrawing a registration leaves the earlier `REGISTRATION_CONFIRMED` notification intact, so the inbox stays an accurate history

9. Unit Testing with pytest

The automated suite runs against a disposable PostgreSQL schema seeded by fixtures, so every run starts from a known state and no test depends on the ordering of another. Email delivery in the password-recovery tests is captured by MailHog rather than sent, so the suite can assert on what would have been delivered without touching a real mailbox.

9.1 Suite Layout – files added or changed in Sprint 2

File	Covers	Author
tests/integration/test_certificate.py	Certificate wallet, time-limited download URL and public serial verification (IT-084 – IT-098)	Pawan
tests/integration/test_password_reset.py	Forgot-password and reset-password flows, including token reuse, expiry and cross-type misuse (IT-415 – IT-426)	Pawan
tests/integration/test_event.py (updated)	Result declaration rework and the post-declaration freeze (IT-275 – IT-286)	Pawan
tests/integration/test_notification.py (updated)	Regression coverage for notifications raised by the reworked result declaration (IT-378 – IT-414)	Pawan
tests/unit/core/test_certificate.py	Serial derivation, text clipping and PDF rendering (UT-427 – UT-444)	Pawan
tests/unit/core/test_mailer.py	Reset-email composition and dispatch (UT-451 – UT-452)	Pawan
tests/unit/core/test_token.py	Access, refresh and reset token creation and decoding (UT-453 – UT-459)	Pawan
tests/unit/utlis/test_hashing.py	Password hashing, verification and fingerprinting (UT-445 – UT-450)	Pawan

9.2 Commands

```
$ pytest backend/tests -q # full cumulative suite
$ pytest tests/integration/test_certificate.py -v
$ pytest tests/integration/test_password_reset.py -v
$ pytest backend/tests --cov=app --cov-report=term-missing
```

9.3 Results – full cumulative suite

Environment: Python 3.13.3, pytest 9.1.1, pluggy 1.6.0, anyio 4.14.1, pytest-asyncio 1.4.0. The full run completes in 240.34 s.

Module	Test file	Cases	Passed	Failed
Communication – announcements	tests/integration/test_announcement.py	56	56	0
Identity & Access – auth	tests/integration/test_auth.py	27	27	0
Certificate	tests/integration/test_certificate.py	15	15	0
Club Management	tests/integration/test_club.py	88	88	0

Module	Test file	Cases	Passed	Failed
Identity & Access - onboarding	tests/integration/test_college.py	22	22	0
Event Management	tests/integration/test_event.py	78	78	0
Communication - issues	tests/integration/test_issue.py	66	66	0
Membership	tests/integration/test_membership.py	25	25	0
Notifications	tests/integration/test_notification.py	37	37	0
Identity & Access - password recovery	tests/integration/test_password_reset.py	12	12	0
Integration subtotal (IT-001 - IT-426)	-	426	426	0
Certificate service (unit)	tests/unit/core/test_certificate.py	18	18	0
Mailer (unit)	tests/unit/core/test_mailer.py	2	2	0
Tokens (unit)	tests/unit/core/test_token.py	7	7	0
Password hashing (unit)	tests/unit/utills/test_hashing.py	6	6	0
Unit subtotal (UT-427 - UT-459)	-	33	33	0
Total	-	459	459	0

```

===== test session starts =====
platform linux -- Python 3.13.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /mnt/C258673838673299/PROJECT/software_engineering_project_code/MAY2026-Team-003/backend
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.14.1, asyncio-1.4.0, json-report-1.5.0, metadata-3.1.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 459 items

tests/integration/test_announcement.py ..... [ 12%]
tests/integration/test_auth.py ..... [ 18%]
tests/integration/test_certificate.py ..... [ 21%]
tests/integration/test_club.py ..... [ 40%]
tests/integration/test_college.py ..... [ 45%]
tests/integration/test_event.py ..... [ 46%]
tests/integration/test_issue.py ..... [ 62%]
tests/integration/test_membership.py ..... [ 74%]
tests/integration/test_notification.py ..... [ 82%]
tests/integration/test_password_reset.py ..... [ 90%]
tests/unit/core/test_certificate.py ..... [ 92%]
tests/unit/core/test_mailer.py .. [ 96%]
tests/unit/core/test_token.py ..... [ 97%]
tests/unit/utills/test_hashing.py ..... [100%]

===== 459 passed, 3 warnings in 240.34s (0:04:00) =====
pawan@Pawan: .../MAY2026-Team-003/backend$

```

Figure 7: pytest run of the cumulative suite at the end of Sprint 2 - 459 collected, 459 passed

10. Defects Found – Expected vs Actual

The course asks for at least one case where the actual output differed from the expected output. This section reports the Sprint 2 position honestly, including the part of it that is less flattering than Milestone 3.

10.1 Sprint 2 Functional Defects

No functional defect was found during Sprint 2 testing: all 459 cases in the cumulative suite passed on the run recorded in §9.3, including the 60 added this sprint. The team has chosen to report that as it stands rather than manufacture a failing case, and to be explicit about why the number is not evidence of a defect-free system.

- The four defects logged in Milestone 3 (BUG-01 to BUG-04) were caught precisely where behaviour crossed a boundary between modules – capacity under concurrent registration, a registration read without its event, a club read without its college scope, a state transition without a guard. Sprint 2 added a module (Certificate) that is reached through a single well-defined trigger and has one owner, which is the shape of code least likely to produce that class of defect.
- The modules with the heaviest coupling – attendance, results, certificates and the leaderboard – are backend-complete but not yet integrated with the frontend (§11). The defects that surfaced in Sprint 1 were overwhelmingly integration defects, and that integration has not happened yet for these modules. The team expects that class of defect to appear during Milestone 5, and has scheduled the remaining integration early enough in that milestone to absorb it.
- A pass rate of 459/459 says the implementation matches the test suite. It does not say the test suite is complete. The suite currently asserts the paths the team predicted; the Milestone 5 plan therefore keeps testing capacity in reserve rather than treating the suite as finished.
- The figure covers the backend suite only. The frontend suite is currently red on the dev branch and is tracked as issue #56 – see §10.3.

10.2 Documentation Defect Carried Open from Milestone 3

DOC-01 remains the one recorded case in this milestone where the delivered artefact does not match what was expected of it.

ID	Artefact	Expected	Actual	Status
DOC-01	campus-connect-open api.yaml – all 50 operations	The specification declares 401, 403, 404 and 409 alongside 200 and 422, and each operation description carries its user story id	Only 200 and 422 are declared. Error paths are raised as HTTPException without a responses= mapping on the route decorator, so FastAPI cannot infer them. Descriptions carrying a	Open – carried to Milestone 5, owned by Shrestha (feedback item F-05)

ID	Artefact	Expected	Actual	Status
			story id are present on two operations only	

Root cause: route decorators do not carry a responses= argument or a description. Fix: add responses={401: ..., 403: ..., 404: ..., 409: ...} and a description naming the story to every route decorator, then re-export the YAML. Re-verification: the exported specification must declare, for each protected operation, every status code that the corresponding test cases in §8 already assert against the deployed build.

The distinction matters and the team wants it on the record: the runtime behaviour is correct and tested – 401, 403, 404 and 409 are asserted throughout §8 and pass. What is defective is the document that describes the behaviour, not the behaviour. Reporting it as closed would have been the easier option and would have been false.

10.3 Open Issues at the Close of Sprint 2

Seven issues were closed during Sprint 2 and three remain open on the dev branch. They are listed here rather than left to the screenshot, because one of them is a genuine expected-versus-actual case even though it sits outside the API suite.

Issue	Title	Position at submission
#56	Frontend test suite is red on dev – 35 of 60 failing	Open. The frontend suite was written against the mocked data layer and has not been updated for the live endpoints wired this sprint, so the failures are stale expectations rather than broken screens – the five Live modules in §11 were validated manually against the deployed build. Realigning the suite is scheduled with the remaining integration work in Milestone 5.
#54	Endpoint to view another member's profile	Open, labelled backend / bug / enhancement. Raised from the frontend side; the club roster returns member names but not a viewable profile. Scheduled for Milestone 5.
#50	Backend endpoints required to complete UI (follow-up to #46)	Open, 4 of 6 sub-items complete. The remaining two are the endpoints the un-integrated screens in §11 depend on.

Issue #56 is the honest counterpart to the 459/459 in §9.3. The API suite is green; the frontend suite is not. The team is reporting both rather than the flattering half, and the gap between them is exactly the integration work §11 and §13 schedule for Milestone 5.

All issues

New issue

is:issue state:open

☐ Open

3

Closed

7

Author ▾ Labels ▾ Projects ▾ Milestones ▾ Assignees ▾ 🔼🔽 Newest ▾

☐  Frontend test suite is red on dev : 35 of 60 failing

#56 · 1mystic opened 3d ago



☐  Endpoint to view another member's profile

backend

bug

enhancement

#54 · 1mystic opened 3d ago

☐  Backend endpoints required to complete UI (follow-up to #46)

 4 / 6

backend

#50 · 1mystic opened 3d ago



Figure 8: Issue tracking in GitHub at the close of Sprint 2 – 3 open, 7 closed

11. Frontend and Backend Integration

The frontend is deployed at <https://campus-connect-swe.vercel.app/> and the backend at <https://campusconnect.itshrestha.dev>. Integration is reported by module and by state, and the states are used strictly: a module is Live only if the deployed frontend calls the deployed endpoints for every screen in that module.

Module	State	Detail
Authentication	Live	Signup, login, token storage and refresh, route guards, and the password-recovery screens added this sprint
College Onboarding	Live	Campus-admin onboarding wired to POST /college/onboarding
Clubs	Live	Directory, search and filters, club profile, join request, my clubs, leader request queue, admin approval queue
Events	Live	Event feed and detail, creation and publish, registration and withdrawal, my registrations
Announcements	Live	Consolidated feed, composer, pinning, unread badge
Notifications	In progress	Inbox and unread badge partially wired; completion is the first item of Milestone 5
Attendance	Backend complete, integration pending	Endpoints delivered and tested; console not yet wired
Results	Backend complete, integration pending	Single-call declaration endpoint delivered and tested; leader console not yet wired
Certificates	Backend complete, integration pending	Wallet, download and public verification delivered and tested; student profile view and the public verification page not yet wired
Leaderboard	Not implemented	Story 7.1 carried to Milestone 5
AI Club Finder	Rebuilt, integration pending	Redesigned as an agentic module after client and TA feedback; complete and tested in isolation, integration scheduled for Milestone 5

The four backend-complete modules are ready independently and could have been wired quickly at the end of Sprint 2. They were held back deliberately: attendance, results and certificates form one chain in which a change at any point moves the state of the other two, and a hurried integration of that chain in the final days of the sprint could not have been validated to the standard the rest of the system has been held to. They will be completed carefully before Milestone 5. The same reasoning applies to the AI module, whose architecture changed after the Sprint 1 feedback

12. User Feedback from the Sprint 2 Demo

The Sprint 2 build was demonstrated to the same primary and secondary users interviewed in Milestone 1 and shown the Sprint 1 build, so feedback is continuous across the project rather than gathered from a fresh audience each time. Each session walked through what changed since the last demo – password recovery, the certificate on the student profile, and the reworked result declaration – followed by unscripted use of the deployed build. Where a user's Sprint 1 request was delivered this sprint, we asked specifically whether the change solved the problem they originally described.

12.1 Student – Shikha Singh

"The app has improved a lot – the forgot password feature was something I really needed, and it's great to see it working now."

- Sprint 1 ask: forgot password, an event reminder ahead of the start, the certificate alongside the result, and interest-based club search.
- Sprint 2 response: password recovery delivered (F-06); event reminder notification delivered (F-03); certificate engine delivered on the backend with the profile view still to be wired; interest-based search carried to Milestone 5.
- Confirmed the recovery flow solved the problem she described – she reset a password end to end during the session.
- Noted the interest-based club search is still not available and is what she would reach for first when looking for a club.

12.2 Club Leader – Aayansh Yadav

"The result workflow feels much more natural now – it's closer to how we actually run events."

- Sprint 1 ask: bulk check-in and a searchable roster; default-and-override results.
- Sprint 2 response: result declaration delivered (F-02); bulk check-in carried to Milestone 5.
- Declaring a winner and a runner-up in one action, with everyone else defaulting to participant, matches how an event actually ends and removes the per-student loop he objected to.
- Marking attendance is still one student at a time, and remains his largest remaining friction for a fifty-person event.

12.3 Club Ambassador – Aayush

"Certificates going out automatically is the part that saves us the most work – that used to be a whole evening."

- Sprint 1 ask: a seen count on an announcement for the leader dashboard.

- Sprint 2 response: not delivered. The seen count is carried to Milestone 5 as a schema change, since it needs per-member read tracking rather than a single seen-at marker.
- Valued that a certificate is generated without anyone remembering to do it, and that the QR code resolves to a public verification page a recruiter could use.
- Still cannot tell how many members have read a post, which was his original request.

12.4 Campus Administration

"The platform continues to feel polished and well-structured - the team has clearly been responsive to the feedback."

- Sprint 1 ask: a visible signal of which clubs are actually active, to inform budget allocation.
- Sprint 2 response: not delivered. The club activity leaderboard (Story 7.1) is carried to Milestone 5.
- Noted that verifiable certificates strengthen the council's case for club funding, because participation becomes auditable rather than self-reported.
- Repeated that budget allocation still has no activity signal to work from, which keeps Story 7.1 the highest-value remaining item from the administration's point of view.

13. Plan for Milestone 5 (Final Submission)

Milestone 5 is due on 23 August 2026. It carries three kinds of work: the two stories carried over from Sprint 2, the remaining frontend integration, and the consolidation and presentation work the course requires for the final submission. Week 12 is reserved by the course for the project.

13.1 Carried-Over Development Work

ID	Item	Owner	Note
6.1	AI Club Finder - integrate the agentic module	Atharv	Module complete in isolation; integration held back so it can be validated to the same standard as the rest of the system
7.1	Club activity leaderboard	Shrestha	Weighted activity score over events run, members recruited and participation recorded
-	Frontend integration of Notifications, Attendance, Results and Certificates	Shrishti, Pawan	The highest-risk remaining work and the most likely source of defects; scheduled first in the milestone so failures have time to be fixed
F-01	Bulk check-in and searchable participant roster	Pawan, Shrishti	Requested twice by the same user; scheduled with the attendance console integration
F-04	Per-announcement seen count for the leader dashboard	Pawan	Requires per-member read tracking; scheduled if capacity allows
F-05	DOC-01 - declare 401 / 403 / 404 / 409 and story ids on all 50 route decorators, re-export the YAML	Shrestha	Mechanical; re-verified against the status codes already asserted in §8
#56	Realign the frontend test suite with the live endpoints (35 of 60 currently failing)	Shrishti	Suite still asserts against the mocked data layer replaced this sprint
#54	Endpoint to view another member's profile	Shrestha	Raised from the frontend side during integration

13.2 Submission and Presentation Work

Item	Owner	Target
Recorded video demo of the working system	Atharv, Shrishti	Before 23 August 2026
Presentation slides - project overview, user identification, user stories	Atharv, Kavisha	Before 23 August 2026
Final consolidated report compiling Milestones 1-5, tools and technologies, and issue-tracking evidence	Kavisha (assembly), all (sections)	23 August 2026

Item	Owner	Target
Complete code zip with README and run instructions (Ubuntu / macOS and Windows)	Shrestha, Shrishti	23 August 2026
Code review and issue-tracking evidence section with screenshots	Shrestha (code review), Kavisha (issue tracking)	23 August 2026
Future-scope list with documented deferral reasons (Google authentication)	Atharv	23 August 2026
Client review of the complete build	Atharv	Client window from 16 August 2026

The graded project showcase is scheduled for 28 August 2026, 6:30 – 7:00 PM. The demo flow will be confirmed after the client review.

14. Submission Artefacts

Artefact	File / Location	Owner
This report	Milestone4DeliverablesTeam003Nexmind.pdf	Atharv, Kavisha
Swagger-compatible API documentation	campus-connect-openapi.yaml	Shrestha
Backend source code	backend.zip (app/, alembic/, requirements.txt, README.md)	Shrestha
pytest suite	backend/tests/ inside backend.zip	Pawan
Full test-case matrix (459 cases)	Linked in §8; also in backend/tests/test-cases/	Pawan, Kavisha
Live API	https://campusconnect.itshrestha.dev/docs	Shrestha
Live frontend	https://campus-connect-swe.vercel.app/	Shrishti